

Linear Algebra and Optimization

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1 Vectors, Matrices and Linear Equations

Definition 1: Linear Function

A function f that maps n dimensional vectors to m dimensional vectors is called **linear** if it satisfies the properties:

1. For all vectors x and all scalars α , $f(\alpha x) = \alpha f(x)$.
2. For all vectors x and y , $f(x + y) = f(x) + f(y)$

For example, take f to be the function that multiplies the entries of x by 2. In this case, we can easily see f is linear. The challenge arises from more exotic functions, such as swapping 2 coordinates of x , or adding one coordinate to all the others. It is important to note that **every linear function can be represented as a matrix**.

Definition 2: Matrix-vector product

The matrix-vector product of a 2×2 matrix and a 2 dimensional vector is defined as

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

We may also interpret this product as a linear combination of the columns of the matrix, which makes it easier to work with higher dimensions.

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} a \\ d \\ g \end{bmatrix} + y \begin{bmatrix} b \\ e \\ h \end{bmatrix} + z \begin{bmatrix} c \\ f \\ i \end{bmatrix}$$

Moreover, we can interpret the result as a column matrix, where each entry is the dot product of the $n \times m$ matrix and our vector.

Example 1

What does the function that corresponds to multiplying a vector by the following matrix do to a point?

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Proof. Take the product of our matrix and a generic 2 coordinate vector.

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -x \end{bmatrix}$$

If we graph this visually with any vector, we see that the function corresponds to rotating every vector in the $x - y$ plane clockwise by $\pi/2$. $\square$

Example 2

What if we want to rotate every vector in the $x - y$ plane counterclockwise by $\pi/4$?

Proof. By far the easiest way to think of these (excluding just knowing it by heart which comes with practice) is to use the standard rotation matrix! The standard matrix for **counterclockwise** rotation is:

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

And now we can just plug in our desired angle. In this case, it's $\pi/4$:

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

□

Theorem 1 : Matrix-Vector Products are Linear Functions

We claim that any matrix-vector product is a linear function. Let's check this.

Proof. Take any scalar α .

$$f\left(\alpha \begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \alpha x \\ \alpha y \end{bmatrix} = \begin{bmatrix} \alpha ax + \alpha by \\ \alpha cx + \alpha dy \end{bmatrix} = \alpha f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$$

And take any two 2 vectors as shown below.

$$f\left(\begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} x' \\ y' \end{bmatrix}\right) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x + x' \\ y + y' \end{bmatrix} = \begin{bmatrix} ax + by + ax' + by' \\ cx + dy + cx' + dy' \end{bmatrix} = f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) + f\left(\begin{bmatrix} x' \\ y' \end{bmatrix}\right)$$

□

Definition 3: General Matrix-Vector Product

If A is an $n \times m$ dimensional matrix and x is an m dimensional vector then Ax is an n dimensional vector and its i th coordinate is

$$(Ax)_i = \sum_{j=1}^m A_{i,j}x_j$$

Theorem 2 : Linear Functions are Matrix-Vector Products

Any linear function f on vectors can be represented as a matrix-vector product. That is, there is some A such that $f(x) = Ax$

Proof. Suppose a vector $x \in \mathbb{R}^m$ and linear function f .

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}$$

We will construct a matrix $A \in \mathbb{R}^{m \times m}$ such that $f(x) = Ax$. We may write x as a linear combination as follows

$$x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \dots + x_m \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

So we have

$$x = \sum_{i=1}^m x_i e_i$$

where e_i are the standard basis vectors in $\mathbb{R}^m$.

$$\begin{aligned} f(x) &= f\left(\sum_{i=1}^m x_i e_i\right) \\ &= \sum_{i=1}^m x_i f(e_i) \end{aligned}$$

Let A be denoted by

$$\begin{aligned} A &= [f(e_1) \quad f(e_2) \quad \dots \quad f(e_m)] \\ &= \begin{bmatrix} a_{11} & \dots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mm} \end{bmatrix} \end{aligned}$$

By the definition of the matrix-vector product we have

$$\begin{aligned} Ax &= \begin{bmatrix} a_{11} & \dots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mm} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \\ &= x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} + \dots + x_m \begin{bmatrix} a_{1m} \\ \vdots \\ a_{mm} \end{bmatrix} \\ &= \sum_{i=1}^m x_i f(e_i) \\ Ax &= f(x) \end{aligned}$$

□

2 Systems of Linear Equations: Geometry and Solving

Definition 1: Inner Product

The inner product between two vectors a and b of dimension n is

$$\langle a, b \rangle = \sum_{i=1}^n a_i b_i$$

More commonly, this is called the dot product and can be written as $a \cdot b$. We can also write this as a matrix-vector product between a $1 \times n$ matrix and an n dimensional vector.

Example 1

Find the 3 by 3 matrix R corresponding to a rotation of 60 degrees around the y axis.

Example 2

Recall that a degree 3 polynomial can be represented with a 4 dimensional vector. Find the matrix corresponding to a derivative in $\mathbb{R}^4$ to $\mathbb{R}^3$ and vice versa for an integral. and also an integral $\mathbb{R}^4$ to $\mathbb{R}^3$.

Suppose we start with two equations: $2x - 3y = 0$ and $x + y = 5$. There are various ways to interpret this.

2.1 Row Interpretation

Each row gives a line.

$$\begin{aligned} \begin{bmatrix} 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= 2x - 3y = 0 \\ \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= x + y = 5 \end{aligned}$$

In this case, we have a solution, since these lines have unequal gradients. This is the expected scenario when the number of variables is equal to the number of constraints. Alternatively, if we have more variables than constraints, we would expect infinitely many solutions. If we had more constraints than variables, we will **likely?** have no solution.

We also have infinitely many solutions if the lines are the same, or the equations always hold true. For example, $0x + 0y = 0$ holds $\forall x, y$

2.2 Column Interpretation

Are there x, y such that

$$x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

It's helpful to graph this on the cartesian plane and see how operations on the vectors affect our visual outcomes.

Theorem 1 : Solving 2x2 Systems

The system of equations

$$\begin{cases} ax + by = \alpha \\ cx + dy = \beta \end{cases}$$

has a unique solution whenever $ad - bc \neq 0$.

Below are three ways to prove this. You can probably tell the one I struggled with...

2.3 Geometric Proof

Proof. The slopes of the lines are $\frac{-a}{b}$ and $\frac{-c}{d}$. The system has exactly one solution when their slopes are unequal, i.e. $ad - bc \neq 0$. $\square$

2.4 Symbolic Proof

Proof. Multiply the first equation by d and the second by b then the second by a and the first by c .

$$\begin{aligned} adx + bdy - bcx - bdy &= d\alpha - b\beta \\ (ad - bc)x &= d\alpha - b\beta \end{aligned}$$

and also

$$(ad - bc)y = a\beta - c\alpha$$

And $x, y \in \mathbb{R}$ iff $ad - bc \neq 0$ $\square$

2.5 Algebraic Proof

Before we get to this proof, it's important to establish a few definitions.

Definition 2: Onto

A function is onto if every element in the codomain is used as an output at least once.

Theorem 2 : Shoelace Theorem

Given a polygon with vertices $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, listed in clockwise order, the area of such polygon is

$$A = \frac{1}{2} |(x_1y_2 + x_2y_3 + \dots + x_ny_1) - (y_1x_2 + \dots + x_ny_1)|$$

Proof. Let

$$\begin{aligned} A &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ B &= \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \end{aligned}$$

We view A as a linear transformation to show that A is both one-to-one and onto. Take a parallelogram with vertices $(0, 0), (a, c), (b, d), (a + b, c + d)$. We can verify by the shoelace theorem that the area of the parallelogram is $|ad - bc|$. Two of the edges of the parallelogram correspond to the two columns of A . Thus, every point inside the parallelogram is some linear combination of the first two columns. Note that we exclude the top and right sides for uniqueness. Since $B \in \mathbb{R}^2$, B is a point within the parallelogram of form

$$\lambda \begin{bmatrix} a \\ c \end{bmatrix} + \mu \begin{bmatrix} b \\ d \end{bmatrix}$$

where $0 \leq \lambda, \mu \leq 1$ for N sufficiently large. We know every point lying within is uniquely expressed as a linear combination of our column matrices (one-to-one and onto). Thus, there is exactly one solution when $ad - bc \neq 0$. $\square$

2.6 More on the Geometry of Linear Equations

Definition 3: Affine Hyperplane

An affine hyperplane is the set of all solutions to a linear equation

$$\langle a, x \rangle = a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

Now the set of solutions to a system of linear equations $Ax = b$ is the intersection of affine hyperplanes.

Theorem 3 : Set of Solutions is the Intersection of Hyperplanes

Let

$$H_i = \{x \text{ s.t. } A_{i,1}x_1 + A_{i,2}x_2 + \dots + A_{i,n}x_n = b_i\}$$

Where H_i is the hyperplane corresponding to the i 'th row of A and b . x is a solution to $Ax = b$ iff

$$x \in \bigcap_{i=1}^m H_i$$

Definition 4: Span of Vectors

The span of vectors $v_1, v_2, \dots, v_n$ is all the vectors that can be obtained as linear combinations of $v_1, v_2, \dots, v_n$.

Moreover, we can deduce:

Theorem 4

If A is $m \times n$, the system of linear equations $Ax = b$ has a solution for every choice of b iff the span of the columns of A is all of $\mathbb{R}^m$.

We can use this theorem to show some pretty interesting results. Consider the system of linear equations

Example 3

Does this system of linear equations have any solutions? If so how many?

$$\begin{bmatrix} 2 & -3 & 0 \\ 1 & 1 & 2 \\ 3 & -2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

Proof. None of the rows of A are scalar multiples of each other, but we can add the first two equations to get an important constraint

$$3x - 2y + 2z = 2$$

which actually contradicts our third equation, so there is no solution for this system. Because we can express the third equation as a linear combination of the first two columns, the span of the columns of A cannot be all of $\mathbb{R}^3$. $\square$

Definition 5: Linear Independence

A set of vectors $v_1, v_2, \dots, v_n$ is linearly independent if no v_i can be expressed as a linear combination of the others. If A is $n \times n$, the system of linear equations $Ax = b$ has a solution $\forall b$ iff the columns of A are linearly independent.

3 Gaussian Elimination

Definition 1: Augmented Matrix

The augmented matrix of a system of linear equations $Ax = b$ is of the form

$$[A|b]$$

Definition 2: Pivot

The pivot in a row is the leftmost nonzero.

The key of Gaussian elimination is that we need 3 invertible operations:

- scale a row by some nonzero number
- add a multiple of a row to another row
- swap rows

Our goal is to have our pivots left to right strictly. That is in **Row Echelon Form**.

Definition 3: Singular Matrix

A matrix A is singular if its row echelon form has a row full of zeroes. This means there is either no solution or an infinite number of solutions.

A variable with no pivot is free. That is, it can be set to anything. The number of dimensions of our solution space is equal to the number of free variables! This is a very cool and important fact to remember!!!!

4 Products, Adjacency Matrices, Matrix Exponentiation

Example 1

$$x_1 + x_2 = 12 \quad (1)$$

$$x_1 = x_3 + x_4 \quad (2)$$

$$x_2 + x_3 = x_5 \quad (3)$$

$$x_4 + x_5 = 12 \quad (4)$$

Proof. We begin with the augmented matrix.

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 12 \\ 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 12 \end{bmatrix}$$

And after changing to row echelon form (REF) we see that the last row is singular (filled with zeroes). The singularity of this matrix leads to an infinite number of solutions (courtesy of our free variables). $\square$

4.1 Matrix Multiplication

$A(Bv) = Cv \implies C = AB$ Suppose we have an $m \times n$ matrix A and $n \times p$ matrix B . Then

$$\begin{aligned} C_{i,k} &= \sum_{j=1}^n A_{i,j} B_{j,k} \\ &= \text{ithrow } A * \text{kthcol } B \\ &= C_k = AB_k \end{aligned}$$

The last line should be understood as the k th column of C .

Theorem 1 : Properties of Matrix Multiplication

We have the associative property such that $A(BC) = (AB)C$. We have the distributive property (which holds for the right and left) $A(B + C) = AB + AC$. It is important to note that it is not necessarily true that $AB = BA$. In fact, it is more often true that $AB \neq BA$.

Definition 1: Walk

A walk is a sequence of vertices such that every consecutive pair is connected by an edge, with repeats permitted. We can use an adjacency matrix to represent a directed (or undirected) graph of several nodes. Recall Markov Chains from 18.600!

Theorem 2 : Number of n length walks

Suppose an adjacency matrix A . A^n gives the number of n length walks that exist from one vertex to another (or itself) for each vertex. $(A^n)_{i,j}$ is the n length walk from i to j .

Proof. TODO: write a proof for the preceding theorem We induct on n . We first prove the simplest case that A^2 gives the number of length 2 walks between vertices for each vertex. $\square$

Definition 2: Transpose

The transpose of an $m \times n$ matrix A denoted by A^T is an $n \times m$ matrix with

$$(A^T)_{i,j} = A_{j,i}$$

If $A = A^T$ then we say that A is symmetric. We can also express the inner-product using the transpose of a vector

$$\langle a, b \rangle = a^T b$$

Definition 3: Outer Product

The outer product between two n dimensional vectors a and b is an $n \times n$ matrix ab^T . This gives us another interpretation of matrix multiplication. If A and B are $m \times n$ and $p \times n$ matrices with columns $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ respectively then

$$AB^T = \sum_{i=1}^n a_i b_i^T$$

It becomes easier to see when we think of it visually. Take a_i to be an $m \times 1$ column vector. Then take b_i^T to be a $1 \times p$ row vector. Then each multiplication of these vectors produces an $m \times p$ matrix. We then sum all these resulting matrices to get our final answer.

4.2 Inverse of a Matrix

Definition 4: Right Inverse of a Matrix

Given a square matrix A , its right inverse is a matrix A^{-1} with the property that

$$AA^{-1} = I$$

While matrix multiplication is in general not commutative, the left and right inverse of a matrix are always the same

Definition 5: Left Inverse of a Matrix

A matrix A^{-1} is a left inverse, i.e. $A^{-1}A = I$ if and only if it is the right inverse, i.e. $AA^{-1} = I$.

Suppose we want to solve a linear system $Ax = b$. If we know the inverse A^{-1} , we can get a closed form expression for the solution. Consider $x = A^{-1}b$.

$$Ax = A(A^{-1}b) = (AA^{-1})b = Ib = b$$

5 Subspaces and Nullspaces

Definition 1: Span

The span of $v_1, v_2, \dots, v_n$ where $v_i \in \mathbb{R}^m$ is all linear combinations in $\mathbb{R}^m$ that can be formed of all n vectors.

Definition 2: Vector Space

A vectors space is a set V of objects closed under addition and scalar multiplication (closed under linear combinations). That is $\forall v, w \in V, v + w \in V$. $\forall v \in V, \alpha \in \mathbb{R}, \alpha v \in V$

Definition 3: Subspace

A subspace $S \subseteq V$ is also closed ... (similar to above). A set of vectors $v_1, \dots, v_\lambda$ generate S if $\forall v_i \in S$ and $\text{span } v_1, \dots, v_\lambda = S$

Theorem 1 : Subspaces

For subspaces S_1 and S_2 , $S_1 \cap S_2$ is a subspace. $S_1 \cup S_2$ is not necessarily a subspace.

Definition 4: Column Space

Given a matrix A , the column space of A is the span of its columns. That is for $m \times n$ matrix A we have

$$\begin{bmatrix} a_1 & a_2 & \dots & a_n \end{bmatrix}$$

$$C(A) = \text{span}(a_1, \dots, a_n) = \{Ac \mid c \in \mathbb{R}^n\}$$

Theorem 2

$C(A), N(A)$ are vector spaces ($N(A)$ is the nullspace of A).

Theorem 3

For $A_{m \times n}$, $C(A) \subseteq \mathbb{R}^m$, $N(A) \subseteq \mathbb{R}^n$ and vice versa A^T

Definition 5: Nullspace

The nullspace of A is the set of all vectors x that make the equation $Ax = 0$ true.

Theorem 4

Consider $Ax = b$.

If $b \notin C(A)$ then no solution. If $b \in C(A)$ then we have to consider two cases:

- $N(A) = \{0\} \implies 1$ solution.
- $N(A) = \{0, x_1, \dots\} \implies$ infinite solutions.

Theorem 5 : Long ass Theorem

For $A_{n \times n}$, A invertible $\iff C(A) = \mathbb{R}^n \iff$ cols of A are linear independent $\iff N(A) = \{0\} \iff Ax = b$ uniquely solvable $\forall b$.

Theorem 6 : Important theorem

Gauss-Jordan elimination preserves nullspace at every step. It is important to note that Gauss-Jordan does NOT preserve the column space of A .

For B invertible and $A = BA'$ where A' is some matrix, $N(A') = N(A)$. If B not invertible $N(A') \leq N(A)$

Theorem 7

For T invertible, $N(A) = N(AT)$

$x \in N(TA) \iff TAx = 0 \iff T^{-1}TAx = T^{-1}0 \iff Ax = 0$

Theorem 8

For $A_{n \times n}, B_{n \times n}$, if $x \in N(A)$ and $x \in N(B)$, $x \in N(A + B)$

Theorem 9

$C(AB) \subseteq C(A)$